Final Project

🚀 Final Project: Industry-Simulated AI Product Workflow

AI Development & Collaboration Project

Using CrewAI, GitHub, Flask, Streamlit

1. Business Background

A retail-tech company is scaling its data and AI capabilities. They collect large volumes of sales, customer, and product data and want to operationalize AI models in a reproducible, team-friendly way.

Management requires two distinct perspectives:

1. Business & Descriptive Understanding
   → "What has happened in the business?"
2. Predictive Modeling
   → "What is likely to happen next?"

Your task is to simulate how real AI product teams collaborate in industry to turn data into deployable insight and predictive intelligence.

2. High-Level Mission

Your team (up to 5 students) must design and implement a CrewAI Flow coordinating two separate multi-agent teams:

👥 Crew 1 — Data Analyst Crew (at least 3 agents)

Responsibilities

- Ingest and validate a dataset chosen freely (Kaggle or public open data).
- Clean, preprocess, and document the dataset.
- Perform descriptive analytics and exploratory data analysis.
- Produce visual insights and business summaries.
- Define a dataset contract that the Data Scientist crew must follow.

Required Outputs

- clean_data.csv
- eda_report.html
- insights.md

- dataset_contract.json
  (schema, allowed values, assumptions, constraints)

👥 Crew 2 — Data Scientist Crew (at least 3 agents)

Responsibilities

- Read and validate the cleaned dataset + the dataset contract.
- Perform feature engineering.
- Train at least one predictive model.
- Evaluate comparisons between at least 2 model variations.
- Produce a model card describing:
    - Model purpose
    - Training data summary
    - Metrics
    - Limitations
    - Ethical considerations

Required Outputs

- features.csv
- model.pkl (or .joblib)
- evaluation_report.md
- model_card.md

🔗 CrewAI Flow Requirements

Your Flow must:

1. **Automate handoff from Data Analyst Crew → Data Scientist Crew.**
2. **Include validation steps, such as:**
    - Confirm the dataset_contract matches the cleaned dataset.
    - Confirm all required features exist before modeling.
3. **Support reproducibility:**
    - Deterministic steps
    - Clear logging
    - Artifacts saved inside the repo

4. Fail gracefully when validations break.

🏗 Required Tech Stack

Your team must use:

Core Requirements

✔ **CrewAI**
✔ **Python**
✔ **Git + GitHub (with Pull Requests)**
✔ **Streamlit or Flask (or both)**
✔ **Pandas, Scikit-Learn, Matplotlib/Seaborn**

Optional Deployment (recommended):

- **Streamlit Cloud**
- **Railway**

Alternative Path

If your group does not want to build the connected Flow, you may submit a **1-page project proposal** (problem, dataset, method, expected output). This must be approved before you begin.

Final Deliverables

- Repository source code, artifacts, and documentation.
- Business presentation (10–12 slides) or demo.
- Short demo video (≤5 min).